

**CS 4850-02 • Spring 2026 • January 25, 2026**

Professor Sharon Perry

PROJECT **SP-5Red**

# **Grocery List App**

*Final Report*

**276**  
Total Man Hours

**3874 (3613 Dart  
and 261 SQL)**

**7**  
Project  
Components

**Joshua Dupati**  
Developer (Team Lead)

**Ash Arul**  
Documentation / Testing

**Jacob Brockel**  
Developer / Documentation

Website: <https://happyshoppercs4850.github.io/HappyShoperWebsite/index.html>

GitHub: <https://github.com/HappyShopperCS4850/HappyShoperApp>

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# 1. Introduction

## 1.1 Project Overview

This project focuses on the design and development of a cross platform grocery list application. The application will support user accounts and shared grocery lists. Multiple users will be able to view and update the same list in real time. All changes will sync across devices to keep data consistent.

The system will include user authentication and a centralized database for storing user and list data. The project scope includes requirements analysis, database design, interface mockups, application development, testing and deployment of a working prototype. The project follows standard software development life cycle practices and emphasizes collaboration, version control and testing.

## 1.2 Platform

Grocery List app development using Flutter and Dart. Flutter is an open-source UI SDK that allows for development of both iOS and Android apps. It is a Google product that works with the Dart programming language which is easy to learn and provides natively compiled, optimized applications.

## 1.3 Collaboration Tools

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Communication</b> | Teams, iOS Group Chat, Google Workspace |
| <b>Collaboration</b> | GitHub, Google Workspace, Slack         |

## 1.4 Statement of Participation

By submitting this assignment, I acknowledge that I will participate in all meetings, communications, deliverables and other tasks necessary to complete the project. If I do not participate, I understand that Professor Perry will meet with me to remedy the situation.

## 1.5 Definitions and Acronyms

|             |                                     |
|-------------|-------------------------------------|
| <b>SRS</b>  | Software Requirements Specification |
| <b>SDD</b>  | Software Design Document            |
| <b>CRUD</b> | Create, Read, Update, Delete        |
| <b>UI</b>   | User Interface                      |
| <b>API</b>  | Application Programming Interface   |

## 2. Requirements

### 2.1 Project Scope

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The scope of this project includes requirements analysis, system design, interface mockups, application development, testing, and deployment of a working prototype. The application will support user authentication, shared grocery lists, real time synchronization, and basic list management functionality.

### 2.2 Functional Requirements

#### 2.2.1 User Account Management

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- The system shall allow users to create an account using an email and password
- The system shall allow users to log in using valid credentials
- The system shall prevent unauthorized access to user data

#### 2.2.2 Authentication

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- The system shall authenticate users before granting access to grocery lists
- The system shall maintain a logged in user until logout

#### 2.2.3 Grocery List Management

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- The system shall allow users to create a new grocery list
- The system shall allow users to add items to a grocery list
- The system shall allow users to update grocery list items
- The system shall allow users to delete grocery list items

#### 2.2.4 Shared Lists

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- The system shall allow users to invite other users to a grocery list
- The system shall allow multiple users to view and edit the same grocery list
- The system shall synchronize grocery list updates in real time across all users

#### 2.2.5 Navigation

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- The system shall display a home screen listing all grocery lists associated with the user
- The system shall allow users to navigate between lists and application screens

### 2.3 Non-Functional Requirements

#### 2.3.1 Security

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- The system shall securely store user authentication credentials
- The system shall restrict access to grocery lists to authorized users only

#### 2.3.2 Capacity

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- The system shall support multiple users accessing the same grocery list simultaneously

### **2.3.3 Usability**

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- The application shall provide a simple and intuitive UI
- The application shall be usable by non-technical users

### **2.3.4 Performance**

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- Grocery list updates shall be reflected to all users within a reasonable time frame

## **2.4 External Interface Requirements**

### **2.4.1 User Interface**

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- The application shall provide mobile-friendly screens optimized for touch interaction

### **2.4.2 Hardware Interface**

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- The application shall run on standard iOS and Android mobile devices

### **2.4.3 Software Interface**

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- The application shall interface with a backend database for data storage and synchronization

### **2.4.4 Communication Interface**

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- The application shall use network communications to support real time data

## **2.5 Constraints**

### **2.5.1 Environment**

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- The application will be developed using Flutter / Dart
- The application must run on both iOS and Android platforms

### **2.5.2 System**

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- The application will rely on a centralized backend database to store user data and grocery lists
- Real time synchronization depends on network availability
- Must support both iOS and Android
- Time constraint is confined to the semester
- List editing is confined to authorized members
- Storage limitations of Supabase backend

## 3. Analysis / Design

### 3.1 Use Cases

---

- User creates an account and logs in
- User creates a grocery list
- User shares a grocery list with another user
- Multiple users update a shared grocery list simultaneously

### 3.2 Data Flow

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Users' actions in the application are sent to the backend database, which processes and stores the updates. Updated data is then synchronized and displayed across all connected user devices in real time.

### 3.3 Design Considerations

#### 3.3.1 Assumptions and Dependencies

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- Users have an iOS or Android device
- Internet connectivity is available
- User authentication via email and password
- Users trust each other for list collaboration
- Flutter, Dart, PostgreSQL Database

### 3.4 Development Methods

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The software design follows the Agile method of development. The process is iterative in nature and breaks the development process into small manageable segments for continuous delivery. Self-organization, collaboration, and flexibility are the focal points of our team's development strategy.

### 3.5 Architectural Strategies

#### 3.5.1 Cross-Platform Framework

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Cross platform framework will be developed through the use of Flutter and Dart from a single codebase. React Native was considered but the all-in-one self-contained toolkit of the Flutter SDK was chosen.

#### 3.5.2 Backend Architecture

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The open source, back-end-as-a-service platform Supabase was used for backend development. Supabase provides authentication, storage, and real time data synchronization. Connection to Supabase is done through a RESTful API foundation. Supabase streamlined the application and data layer of our 3-tier client server architecture.

### 3.5.3 Database

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PostgreSQL with its relational database format was chosen to store users' accounts and list information. The relational model provides data integrity and clearly identifies connections between users, groups, and list information.

### 3.5.4 Real-Time Synchronization

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List updates between multiple users in a shared list is as close to real time as possible. Supabase monitors changes to the PostgreSQL write ahead log. Once changes are detected updates are broadcasted to subscribed users over websockets.

### 3.5.5 Authentication

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Authentication is ensured using JSON Web Tokens (JWTs). Access tokens confirm user ID in each API request and refresh tokens prevent unwanted signouts. Credentials are located in the system auth table with hashing for further encryption. Row level security ensures users can only read, write, update, or delete appropriate rows.

## 3.6 System Architecture

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The system is organized using a three-tier client-server architecture. The presentation tier and UI is handled by Flutter built using the Dart language. The application tier is handled by Supabase. API connections and authentication are provided and configured within Supabase. The data tier is handled by our PostgreSQL database stored with Supabase.



## 4. Development

### 4.1 Detailed System Design

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Most components described in the System Architecture section will require a more detailed discussion. Other lower-level components and subcomponents may need to be described as well. Each subsection of this section will refer to or contain a detailed description of a system software component.

#### 4.1.1 Classification

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The SP-5Red Grocery List App follows a client-server architecture. The system consists of a mobile front end application and a centralized backend service. The front end handles user interaction while the backend manages data storage, authentication and synchronization.

The application will be developed using Flutter to support both iOS and Android platforms. A backend service will store user accounts, grocery lists and shared list data. Real time updates will be handled through network communication between the client and backend.

#### 4.1.2 Definition

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The system is divided into three main components:

- User Interface Component — allows users to log in, create grocery lists, edit items and share lists with others. This component focuses on usability and clear navigation.
- Application Logic Component — handles validation, list operations and communication with the backend. It processes user actions before sending or receiving data.
- Database Component — stores user credentials, grocery lists and shared list relationships. It ensures data persistence and supports real time synchronization across users.

#### 4.1.3 Constraints

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The application depends on network availability for real time synchronization. Users must have an active internet connection to view and update shared grocery lists. Offline access will be limited and updates may not sync until connectivity is restored.

The system relies on a centralized backend database. Storage limits and backend service availability may affect performance. Authentication is required before any list data can be accessed. Only authorized users may view or modify shared lists.

The application is constrained to mobile platforms supported by Flutter. Device performance and operating system limitations may impact responsiveness.

#### 4.1.4 Resources

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The application uses device resources such as memory, storage and network connectivity. A backend database is used to store user accounts, grocery lists and shared list relationships. A backend authentication service is required to manage user access.

Multiple users may access the same grocery list at the same time. This can create race conditions if updates occur simultaneously. These issues will be handled by backend synchronization logic and controlled write operations to ensure data consistency. Deadlock situations are unlikely due to the stateless nature of client requests.

### 4.1.5 Interface / Exports

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The application provides services for user authentication, grocery list creation, item management and list sharing. These services allow users to create, edit, delete and view grocery lists through the mobile interface.

The system exposes interfaces for retrieving list data, updating items and synchronizing changes with the backend. Each service enforces access control to ensure only authorized users can perform actions. Interfaces are designed to support real time updates and consistent data flow between the client and backend.

### 4.2 Status Contribution — Joshua Dupati

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For Status Meeting 1, Joshua contributed to the preparation of the RAD documentation by developing and organizing the Software Requirements Specification for the SP-5Red Grocery List App project. His work included defining the core functional requirements such as user account management and authentication, grocery list creation and editing, shared list collaboration and navigation flow. He also contributed to outlining system constraints, non-functional requirements, and analysis sections including use case and data flow. He ensured the document structure aligned with the project goal of creating a cross-platform mobile app that supports real-time grocery list synchronization between users.

## 5. Test Plan and Report

### 5.1 Test Plan

The test plan for the SP-5Red Grocery List App was designed to validate functionality across all major system components. The April 17, 2026 milestone included completion of both the Test Plan and Test Report.

### 5.2 Test Cases

| TC #  | Test Case           | Expected Result                                             | Status |
|-------|---------------------|-------------------------------------------------------------|--------|
| TC-01 | User Registration   | User account created successfully with valid email/password | Pass   |
| TC-02 | User Login          | User authenticated and granted access to grocery lists      | Pass   |
| TC-03 | Create Grocery List | New grocery list created and visible on home screen         | Pass   |
| TC-04 | Add Item to List    | Item appears in list immediately after adding               | Pass   |
| TC-05 | Update Item         | Item updated and change reflected in list                   | Pass   |
| TC-06 | Delete Item         | Item removed from list successfully                         | Pass   |
| TC-07 | Share List          | Invited user can view and edit shared list                  | Pass   |
| TC-08 | Real-Time Sync      | Changes reflect across all users simultaneously             | Pass   |
| TC-09 | Unauthorized Access | Non-member blocked from accessing list                      | Pass   |
| TC-10 | Multi-User Editing  | Concurrent edits handled without data loss                  | Pass   |

## 6. Version Control

### 6.1 Version Control Plan

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GitHub will be used as the main version control system for this project. A shared repository will store all source code, documentation and configuration files. Each team member will work on their own branch when developing new features or fixing bugs.

The main branch will contain stable and tested code only. Changes will be merged into the main branch after review and basic testing. Commit messages will clearly describe the changes made. GitHub issues will be used to track tasks, bugs and progress throughout the semester.

### 6.2 Repository

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**GitHub Repository:** [github.com/asharulcoding/sp5-red-grocery-app](https://github.com/asharulcoding/sp5-red-grocery-app)

### 6.3 Risk Assessment

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This project includes several risks that could impact development and delivery.

- Real-Time Synchronization Issues — issues with real time synchronization between users which could cause inconsistent list updates. This risk will be reduced by using reliable backend tools and testing synchronization early.
- Schedule Conflicts — conflicts between team members managed through weekly meetings, clear task assignments and regular check-ins.
- Integration Issues — issues when merging code from multiple contributors. Reduced by using feature branches, regular merges and basic testing before updates reach the main branch.

## 7. Conclusion / Summary

### 7.1 Project Summary

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The SP-5Red Grocery List App is a cross-platform mobile application designed to help users create, manage, and share grocery lists in real time. The project successfully delivered a working prototype that supports user authentication, shared grocery lists, and real-time synchronization across devices.

The team followed Agile development practices throughout the semester, with regular weekly meetings, clear task delegation, and version control through GitHub. The project scope was met within the defined milestones from January through May 2026.

### 7.2 Deliverables Summary

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- Grocery List App available on multiple platforms supporting user accounts and shared grocery lists
- Database for user info and list data
- Individual and group mockups depicting information flow
- Real time list synchronization
- Prototype demonstration
- Test Data and Test Report
- Source Code

### 7.3 References

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- CS 4850 Senior Project Guidelines
- Flutter. Flutter Documentation. Google. <https://docs.flutter.dev/>
- Firebase. Firebase Documentation. Google. <https://firebase.google.com/docs>
- Supabase. <https://supabase.com/docs>
- Dart. <https://dart.dev/guides>
- Balsamiq. <https://balsamiq.com/docs/>
- Canva. <https://www.canva.com/help/>

| Project Name:                                                                        |                                                  |           | SP-5 Red Grocery: Project Plan |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|--------------------------------------------------------------------------------------|--------------------------------------------------|-----------|--------------------------------|------------------|--------------|-------|-------|-------|--------------|-------|-------|-------|--------------|-------|-------|-------|-------|-------|-------|--|--|--|
| Report Date:                                                                         |                                                  |           | 1/24/26                        |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| Phase                                                                                | Tasks                                            | Complete% | Current Status Memo            | Assigned To      |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      |                                                  |           |                                |                  | Milestone #1 |       |       |       | Milestone #2 |       |       |       | Milestone #3 |       |       |       | C-Day |       |       |  |  |  |
|                                                                                      |                                                  |           |                                |                  | 01/25        | 02/01 | 02/08 | 02/15 | 02/22        | 03/01 | 03/08 | 03/15 | 03/22        | 03/29 | 04/05 | 04/12 | 04/19 | 04/26 | 05/03 |  |  |  |
| Requirements                                                                         | Project Selection                                | 0%        |                                |                  | 10           | 6     |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Project Template                                 | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | SRS                                              | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | SOD                                              | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| Project design                                                                       | Database design                                  | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Front End design                                 | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Backend design                                   | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Develop Document                                 | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Test prototype                                   | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Review prototype design                          | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Rework requirements                              | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| Development                                                                          | Document updated design                          | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Test product                                     | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Website                                          | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      | Prototype preparation                            | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| Final report                                                                         | Final report submission to D2L and project owner | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      |                                                  | 0%        |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
|                                                                                      |                                                  |           |                                | Total work hours | 276          | 10    | 11    | 18    | 21           | 17    | 25    | 20    | 16           | 15    | 28    | 25    | 35    | 10    | 25    |  |  |  |
| * formally define how you will develop this project including source code management |                                                  |           |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| <b>Legend</b>                                                                        |                                                  |           |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| Planned                                                                              |                                                  |           |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| Delayed                                                                              |                                                  |           |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |
| Number Work: man hours                                                               |                                                  |           |                                |                  |              |       |       |       |              |       |       |       |              |       |       |       |       |       |       |  |  |  |

| Date         | Milestone                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------|
| Jan 25, 2026 | Project plan defined; work estimate created; required tools and frameworks chosen                |
| Feb 8, 2026  | Functional requirements defined; wireframe mockups; database design; SRS and SDD complete        |
| Mar 8, 2026  | Development documentation; Phase 1 capabilities (user creation, grocery list, member invitation) |
| Mar 17, 2026 | Phase 1 complete; working prototype; presentation; begin Phase 2 (monetization, 3rd party data)  |
| Apr 17, 2026 | Test Plan and Test Report                                                                        |
| May 3, 2026  | Final report package; all app components complete; presentation video; source code               |

**Attendees:** Joshua Dupati, Jacob Brockel  
**Agenda:** Review and analyze completed SRS and SDD sections; delegate remaining items  
**Notes:**

- Discussion between group members occurs daily via iOS group chat

- Meeting times discussed to be moved prior or after class times on Tuesdays or Thursdays
- Action items are being completed ahead of schedule

**Action Items:**

- Section 5, 6, and 7 of SDD to be completed by Ash
- Review and finish formatting for SRS and SDD

**Next Meeting:** *Ensure SRS and SDD are complete prior to February 8th*

**February 3, 2026 — 1:45–2:15 PM — In Person (30 min)**

**Attendees:** Joshua Dupati, Jacob Brockel, Ash Arul

**Agenda:** Review deliverables

**Notes:**

- Reviewed deliverables
- Talked about the app and agreed with the requirements/scope

**Action Items:**

- Look into Flutter
- More research

**Next Meeting:** *Plan the Development phase and start working on the project*

**February 10, 2026 — 1:45–2:15 PM — In Person (10 min)**

**Attendees:** Joshua Dupati, Jacob Brockel, Ash Arul

**Agenda:** Plan Development

**Notes:**

- Josh will work on the front end
- Ash will create mockups

**Action Items:**

- Look into Flutter
- More research
- Plan
- Code

**Next Meeting:** *Check up on the work*

**March 3, 2026 — 1:45–2:15 PM — In Person (10 min)**

**Attendees:** Joshua Dupati, Ash Arul

**Agenda:** Documentation Development

**Notes:**

- Josh will continue to work on the front end
- Jacob will work on the back end
- Everyone will work on the development doc due

**Action Items:**

- Work

**Next Meeting:** *Get ready for the presentation*

#### March 17, 2026 — 1:45–2:15 PM — In Person (10 min)

**Attendees:** Joshua Dupati, Ash Arul, Jacob Brockel

**Agenda:** Presentation

**Notes:**

- Josh added his parts to the presentation of the UI
- Jacob added his parts of the presentation
- Practice

**Action Items:**

- Edit the presentation

**Next Meeting:** *Finishing up the project — Website, App, and turning in the whole portfolio*

#### March 31, 2026 — 2:00–3:00 PM — In Person (1 hr)

**Attendees:** Joshua Dupati, Ash Arul, Jacob Brockel

**Agenda:** Presentation Practice

**Notes:**

- Finalized the presentation
- Assigned speaker roles
- Practiced it

**Action Items:**

- Practiced it

**Next Meeting:** *Sync — Test Cases, Website*



## A.4 Flutter Dart Tutorial


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[Fetch data from the internet](#)
[Add logging for debugging and monitoring](#)

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# Dart tutorial

Begin your Dart learning journey by building an interactive CLI app.

Welcome to the Dart tutorial! This tutorial teaches you how to write Dart code by building an interactive command-line (CLI) application.

You'll start from the very beginning: creating a blank Dart application. By the end, you'll have built an app that demonstrates the essential features of Dart development—variables, functions, classes, async programming, testing, and more!

### Before you begin


**Tip**

**Part of the Flutter learning pathway**

This tutorial is the second step in the [Dart and Flutter learning pathway](#). If you're interested in learning both Dart and Flutter, start there!

This tutorial assumes that you have:

1. A Dart environment set up—if not, follow the instructions to [get the Dart SDK](#).
2. Basic familiarity with programming concepts. The tutorial is beginner friendly, but doesn't cover the fundamentals of programming.

### Lessons

1. [Build your first app](#)
2. [Add interactivity to your app](#)
3. [Write asynchronous code](#)
4. [Organize code with packages and libraries](#)
5. [Define relationships with classes](#)
6. [Handle errors gracefully](#)
7. [Extend your app with enums and extensions](#)
8. [Polish your CLI app](#)
9. [Work with JSON data](#)
10. [Test your app & code](#)
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## A.5 Flutter Dart Tutorial

